Lukas Bozinov

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Programming Skills

- **Languages**: C++, C, Python, C#, JavaScript/TypeScript, Java, Go, SQL
- **Technologies**: AWS, Azure, Kubernetes, Docker, ROCm, PyTorch, OpenCV, Claude Code, Codex, Cursor, AI Agents, Model Context Protocol, PostgreSQL, MongoDB, Linux, Git, CI/CD, CUDA, OpenMP, SIMD

Education

- **Western University** — Graduating April 2027
  *B.S. in Computer Science — GPA: **3.8*** — *London, ON*

Experience

- **AMD (Advanced Micro Devices)** — Markham, ON
  Associate Software Engineer — *May 2026 – August 2026*
  Agentic AI and Machine Learning SWE Intern — *January 2026 – April 2026*
  - **Agentic AI Development**: Created an agentic AI system in Python that autonomously troubleshoots machines with CI pipeline issues using internal tools for intelligent tool calls and problem resolution.
  - **Agent Swarms**: Created specialized agent swarms to accelerate NPU kernel development and bugfixes, increasing efficiency by **35%**. Usable in Claude Code, Codex CLI, and other agentic coding tools.
  - **AI Bug Detector**: Built a KNN model in Python to detect, classify, and correlate bugs in CVML model training pipelines, achieving **87% accuracy** on a test dataset of 15,000+ samples.
  - **Neural Processing Unit (NPU) Testing Application**: Created a tool that pushes NPU utilization to 100% by running multiple ML model training pipelines in parallel, improving test efficiency by **40%**.
  - **Infrastructure Reliability**: Maintained **99% uptime** across 300+ CI testing and CVML model training workstations, ensuring critical development workflows remained operational.
- **J.D. Power** — London, ON (Remote)
  DevOps and MLOps Engineer Intern — *May 2025 – December 2025*
  - **Test Automation & CI/CD**: Built end-to-end automated test suites for web and mobile dashboards using Playwright, integrated with GitLab CI/CD and Xray to enable nightly regression testing; 35% overall reduction in manual testing efforts.
  - **Test Reports**: Implemented Allure reporting & logging, improving defect traceability by **30%**.

Projects & Leadership

- **TensorKernel**: High-performance tensor computation library in C++ featuring optimized kernel implementations (SIMD/OpenMP/CUDA) for neural network operations with built-in profiling and Python/PyTorch interoperability.
- **Chef.ai**: Developed an AI-driven meal planning system in Python using Scikit-learn, Pandas, and NumPy to optimize purchases and reduce food waste by **20%**.
- **Facial Recognition Security System**: Implemented a real-time facial recognition system in C++ using OpenCV and Qt5 with role-based access control and model retraining, achieving **92% accuracy**.
- **EasyConvert**: An Rust-based file converter that converts any format to any other (compatible) format.
- **Track Isolator**: A local-only music stem separation application built in C# and TypeScript, that uses machine learning (Demucs) to isolate vocals and instruments from music tracks.